Convex optimization for machine learning – TD4

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1 Exercises

Exercise 1. (Projected stochastic gradient)
We consider the following optimization problem

$$\min_{\mathbf{x}\in C} \sum_{i=1}^{n} f_i(\mathbf{x}) \tag{1}$$

where $C = [0,1]^d$ and for all i, $f_i : \mathbb{R}^d \to \mathbb{R}$ is differentiable.

1. We define the convex indicator function of the set C as $\iota_C(x) = \begin{cases} 0 & \text{if } \mathbf{x} \in C \\ +\infty & \text{if } \mathbf{x} \notin C \end{cases}$.
   Show that (1) is equivalent to $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{i=1}^{n} f_i(\mathbf{x}) + \iota_C(\mathbf{x})$.
2. Compute the proximal operator of ι_C
3. Write the proximal stochastic gradient method for the resolution of (1).

Exercise 2. (Optimisation with explicit constraints)
We consider the following optimization problem

$$\min_{\mathbf{x}\in C} f(\mathbf{x}) \tag{2}$$

where $C \subset \mathbb{R}^d$ is a convex set and $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable.

1. We define the convex indicator function of the set C as
   $$\iota_C(\mathbf{x}) = \begin{cases} 0 & \text{if } \mathbf{x} \in C \\ +\infty & \text{if } \mathbf{x} \notin C \end{cases}$$
   Show that (2) is equivalent to
   $$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) + \iota_C(\mathbf{x}) \tag{3}$$
2. Show that for all $\mathbf{x} \in C$, $\partial\iota_C(\mathbf{x}) = \{\mathbf{q} \in \mathbb{R}^n \ : \ \forall \mathbf{y} \in C, \langle \mathbf{q}, \mathbf{y} - \mathbf{x}\rangle \leq 0\}$ and that $\partial\iota_C(\mathbf{x})$ is a cone (it is called the *normal cone* to C at $\mathbf{x}$). Show that for all $\mathbf{x} \notin C$, $\partial\iota_C(\mathbf{x}) = \emptyset$.
3. Show that $\mathbf{x}^*$ is a solution to (3) if and only if $-\nabla f(\mathbf{x}^*) \in \partial\iota_c(\mathbf{x}^*)$.
4. Denote $\mathcal{H}_{\mathbf{w},b} = \{\mathbf{x} \in \mathcal{X} : \langle \mathbf{w}, \mathbf{x}\rangle + b = 0\}$. Compute $\partial\iota_{\mathcal{H}_{\mathbf{w},b}}(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^d$.

   (Hint: Show that we can assume $b = 0$ without loss of generality, and use the fact that $\mathcal{H}_{\mathbf{w},0}$ is a vector space.)
5. Prove that the distance of a point $\mathbf{z}$ to $\mathcal{H}$ is equal to

$$\mathrm{d}(\mathbf{z}, \mathcal{H}_{\mathbf{w},b}) = \min_{\mathbf{x}\in\mathcal{H}_{\mathbf{w},b}} \|\mathbf{x} - \mathbf{z}\|_2 = \frac{|\langle \mathbf{w}, \mathbf{z}\rangle + b|}{\|\mathbf{w}\|_2}\,.$$